

Déjà Vu: Uncanny and Serendipitous Re-encountering of Audiovisual Archives

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Abstract

The advent of the "generative age" marks a pivotal moment in AI evolution, with models like OpenAI's GPT-4 and DALL-E transforming content creation across various fields. These advancements bring unprecedented efficiency and new applications but also present challenges, notably AI hallucinations—factually incorrect or nonsensical outputs due to latent space misinterpretations. This research proposes using AI hallucinations and the digital uncanny to improve archival access through a déjà vu generator. This system connects user experiences with archival video clips, creating immersive installations that evoke familiarity and nostalgia, thus transforming archives into dynamic, memory-evoking interfaces.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Applied computing** → **Arts and humanities**.

Keywords

archival art, creative interface, video retrieval

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1 Introduction

We are on the cusp of what many are calling the "generative age" a period defined by the remarkable capabilities of artificial intelligence (AI) to produce content that rivals human creativity [Sundberg and Holmström 2024]. Generative AI models, such as OpenAI's GPT-4 and DALL-E, are transforming the landscape of content creation across various domains, from literature and art to scientific research and software development. These advancements promise unprecedented efficiencies and novel applications, yet they also introduce significant challenges, chief among them being AI hallucinations [Salvagno et al. 2023].

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AI hallucinations have become a well-known problem in the deployment of generative models. Broadly speaking, AI hallucinations refer to instances where AI systems generate outputs that are factually incorrect, irrelevant, or nonsensical [Emsley 2023]. These fictional details arise from the way AI models interpret and generate data. Unlike human cognition, which is grounded in a complex understanding of context and reality, AI models operate based on patterns learned from vast datasets. When these patterns are misinterpreted or poorly encoded, the resulting output can diverge significantly from the intended meaning.



Figure 1: Conceptual generative rendering of the proposed work Déjà Vu

To understand why hallucinations occur, we need to take a step back and examine the concept of latent space in machine learning. Latent space can be thought of as the "mind" of the machine, where data is represented in a compressed form that captures the essential features and relationships within the data. During the training process, the encoder component of the AI model compresses the input data into this latent space, while the decoder reconstructs the data from these representations. Errors can occur at both stages: the encoder may learn incorrect correlations between different parts of the training data, and the decoder may misinterpret these correlations, leading to erroneous outputs. Two specific aspects of

the decoding process contribute to hallucinations. First, decoders might attend to the wrong parts of the encoded input, generating content that does not align with the source material. Second, the design of the decoding strategy itself, such as the use of the top-k sampling to enhance diversity, can inadvertently increase the likelihood of hallucinations [Dziri et al. 2021; Ji et al. 2023].

The core issue of hallucination lies in the mismatch, or distance, between the real-world materials and their computational representations, creating what is often termed the "digital uncanny" - a phenomenon that 'unsettles and estranges concepts of "self," "affect," "feedback," and "aesthetic experience," forcing us to reflect on our relationships with computational media [Ravetto-Biagioli 2019]. While hallucination is often considered a drawback for most computational applications, its "cousin," the digital uncanny, has emerged as an important aesthetic and cultural mode, inspiring many AI and hallucination-based artistic and design practices. [del Campo and Leach 2022; Zylinska 2020]. Through a vacillation between familiar, conventional and disconcertingly strangeness, these practices emphasise and amplify the uncanniness to cultivate reflective and participatory experiences.

Switching from generated content to the other half of the content world - archival content, the creative usage of such distances is seldom explored. One key issue for archives is that traditional access models, such as search-based interfaces and curated exhibitions, limit the access of archival materials to only "popular" ones due to personal and curatorial bias and leave much of the content "dead" or inaccessible. This results in archives, as repositories of historical and cultural material, being underutilized and incomprehensive in practice, rendering them too one-sided and static. Here raises an intriguing question: can we hinge on the hallucination and uncanny for current computational methods to create a novel access model for archives?

Sharing a similar mechanism with AI hallucination, *déjà vu* — or what some refer to as a hallucinated memory — occurs when we encounter a situation reminiscent of an actual memory that we can't fully recall [Bancaud et al. 1994]. Our brain identifies the similarities between our current experience and a past one, activating similar but not identical memories, leaving us with a sense of familiarity that we can't quite place.

Since archives serve as repositories and catalysts of memory and remembrance, this work draws on these two parallel ideas and proposes a *déjà vu* generator based on archival content. Users begin by inputting a situation or experience using the provided keyboard. The system then retrieves relevant scenes from the archive based on a narrative-enhanced language-video joint embedding space, ensuring that the match is nuanced and not limited to mere visual or textual similarity as traditional retrieval systems would. The clips undergo automated post-processing to create a nostalgic nuance reminiscent of *déjà vu*. These clips are then displayed on a cluster of monitors, with a separate monitor available for browsing the actual archival clips and their related metadata for historical context. Fig 1 indicated a conceptual illustration of the installation for this work.

This generator connects individual experiences (a text description) to approximate memories (archival video clips), thereby providing a more dynamic and engaging way to explore archival content. This approach mimics the mechanism of AI hallucinations,

where familiar elements trigger related, though not identical, memories. Such a system would create a rich, immersive interface that opens up the entirety of the archive to users. This not only enhances accessibility but also fosters a deeper connection with historical and cultural materials, transforming archives from static storage spaces into living, memory-evoking experiences that inspire curiosity and creativity.

2 The ARCHIVE

The Radio Télévision Suisse (RTS) dataset comprises a vast collection of over 500,000 videos from the Swiss public broadcaster. Spanning a diverse spectrum of topics, including news, sports, entertainment, and documentaries, the dataset offers a wealth of video content totalling more than 150,000 hours. As a broadcaster, RTS's dataset is more of a record for production-level information. It provides very vague and inconsistent metadata such as "TypeContenu", "Duration", and a very abstract-level description of what the whole 1-hour programme is about (e.g. a detective film). The level of access granted to the content via surrounding metadata is very limited. Since the purpose of this work is to propose a novel interface for re-encounter archival content, for the ease of this work, a subset of the whole RTS archive is chosen. We select a random sample of videos with "TypeContenu - Films and TV series" to fit the overall theme more. The final dataset used for this work contains around 1,000 videos with a total runtime of around 530 hours. This yields an average runtime for each video of 0.53 hours, with no textual descriptions of the video.

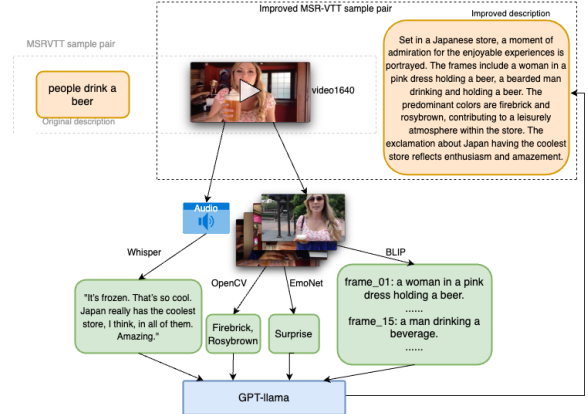


Figure 2: Our overall method for obtaining narrative-enhanced training set

3 Creating the *déjà vu*

Creating the sensation of *déjà vu* with audiovisual archival content fundamentally translates to a task of text-to-video retrieval. The inherent dual distance between the user's input and the retrieved results constitutes the essence of *déjà vu*. Firstly, there is a temporal and spatial gap between the archival content and the user's reality. This is achieved by using archival materials — no matter how accurate the retrieval, it is not the user's actual experience. Secondly, there needs to be an uncanny gap between the retrieved content

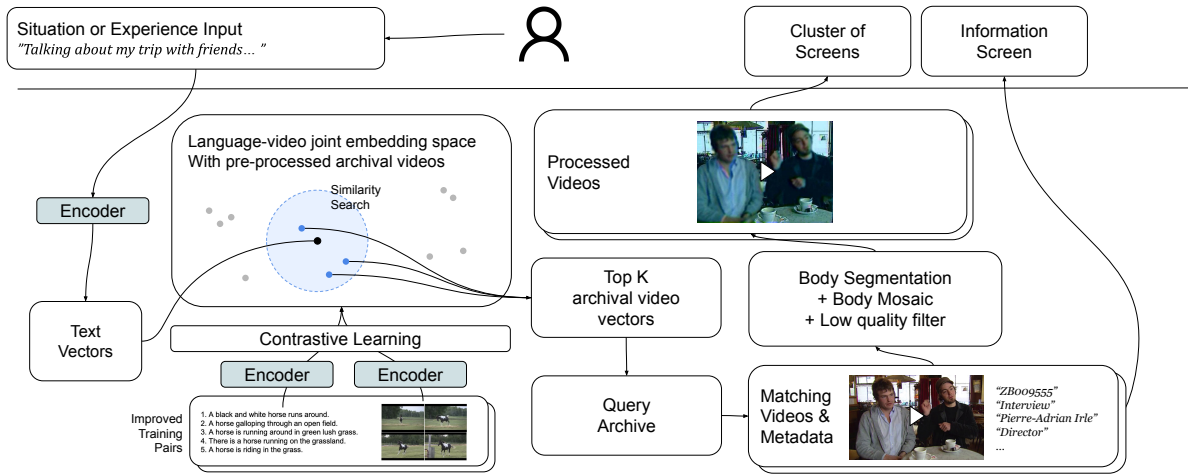


Figure 3: The overall technical workflow for D    Vu. The upper part is the human perspective, and the lower part is the technical perspective.

and the user’s own situation or memories. The challenge lies in delicately balancing familiarity and unfamiliarity during matching, aiming for an as nuanced as possible matching to avoid immediate recognition of the results being “too obvious” or “too contrived” that could diminish the essence of uncanniness for d    vu.

For this, we utilise a pretrained narrative-enhanced language-video joint embedding model as the base. Compared to the traditional single modality-based retrieval (matching only one aspect such as colour, motion, emotion, speech, and visual entities) and a basic text-to-video retrieval (which primarily focuses on visual happenings), the joint embedding space provided by the model directly learns multimodal features from the video data, hence being able to capture a more nuanced and holistic understanding of the scene in detail and abstract level.

Fig 2 indicates our overall method for creating the data for the narrative-enhanced language-video joint embedding model. The main objective is to create better correlation and alignment between the textual and video representations. For this, a novel method to enhance existing language-video datasets was introduced based on the work [Yang 2023], making them more modality and context-aware. This multifaceted video captioning method captures a wide range of elements, including entities, actions, speech transcripts, aesthetics, and emotional cues, ensuring a comprehensive and balanced textual representation of the video content. The improved data is then used to train a space for the retrieval task. Details are illustrated in Section 4.1.1.

4 The Artwork

4.1 Implementation

The overall workflow for the technical part of this artwork is shown in Fig.3. The following paragraphs elaborate on the key technical perspectives, namely how text prompts are used to retrieve archival videos, and how videos are post-processed into “d    vu”.

4.1.1 Text to Video Retrieval. The archival content of this work experiments are detailed in Section 2. The sample dataset is prepared following the base of [Yang and Zhang 2023]. Shot-based segmentations over speech-guided segmentation are used since the archive we work with does not have mature enough transcripts with timestamps. Three shot-based segmentation methods are tested. TransNet V2 [Sou  ek and Loko   2020] is a content-aware detection that works by locating jump cuts, and adaptive content detection. The last two methods are wrapped by the PySceneDetect library and based on OpenCV. We resort to adaptive content detection since it performs the best based on our random sampling and qualitative evaluation. Since the result clips vary in duration and can be too short, a 12-second threshold, as most of the text-video datasets are, is set to group consecutive shot clips into a minimum of 12 seconds.

The core for matching the human inputs to archival video clips is the narrative-aware language-video joint embedding space based on deep learning. Section 3 has already illustrated the method for creating the more nuanced and correlated text-video pairs, that are the foundation for training such a space. An improved version of the classic language-video dataset MSR-VTT [Xu et al. 2016] was created adapting this method as the base to create the new joint-embedding space and model. Following [Gabeur et al. 2020], the text and video in the dataset are encoded into text and video representations and projected to a joint embedding space. This space is then trained to make the distance between the paired text and video as close as possible. The process, in essence, allows the model to learn and align subtle video and textual features directly. All of the RTS videos are then put through the trained model to obtain their embeddings - in other words, their coordinates in the joint embedding space. The input from the user is also transformed into the embedding, and a similarity search is conducted to find the closest archival clips. The retrieved video clips and related metadata are then sent to the information screen for context.

4.1.2 Stylistic Post-processing. “Blurring human figures” and “applying VHS-type video filters” are applied in this post-processing

step. It is to enhance the déjà vu and uncanny effect central to this project. Blurring obscures identities, making the scenes more universally relatable and allowing viewers to project their own memories onto the footage. The VHS filters add a nostalgic, retro aesthetic, evoking a sense of familiarity and temporal dislocation. Together, these techniques amplify the eerie, memory-triggering atmosphere, deepening users' engagement with archival content and reinforcing the exchange between personal and public memories. This approach transforms static archives into dynamic, immersive experiences, aligning perfectly with the project's objectives. For human blurring, Pre-trained TensorFlow.js models¹ provided by Tensorflow is used for body segmentation and mosaic out human body. For the VHS effect, an open-source model ntsc-rs² is used. For the ease of the project, this particular work pre-processed all archival content in use for a more seamless experience. In reality, real-time solutions, especially the VHS effect, would be preferable. One possible solution would be to train a real-time style-transfer model.

Situation or Experience Input
"Talking about my trip with friends..."

Original clips

Post-processed clips

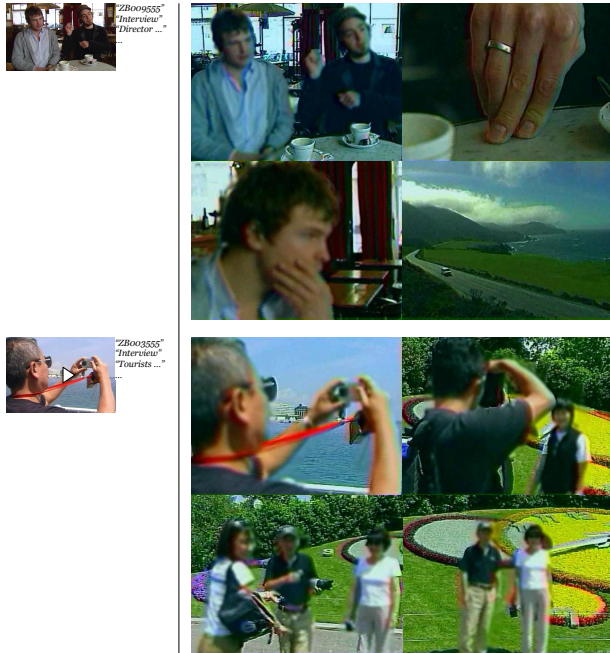


Figure 4: An exemplary overall content sheet for the whole experience, Including text query, retrieved videos and metadata, as well as post-processed videos that will be shown on different screens.

4.2 The Experience

for this specific paper, we choose to use a cluster of old computer monitors as the presentation medium - to emphasise the digital uncanny and nostalgic feeling. Audiences enter any situation or experience description from the keyboard provided, and the system will start to match the archival video clips using the prompt. The post-processed stylistic videos will then be sent to the cluster of monitors to play, while the original videos and their correlated metadata will be sent to an information screen for the context. A detailed exemplary content of the whole experience of the work is given in Fig.4.

5 Conclusion and Future Work

Utilising the improved narrative-aware language-video joint embedding space based on deep learning, déjà vu can 1) retrieve nuanced clips in audiovisual archives using any natural language queries, 2) post-process video clips with various stylistic techniques to emphasise the hallucinated memory and uncanny effect central to this project, and 3) provide serendipitous and natural access and interface to the audiovisual archive and metadata. This innovative approach transforms archives into dynamic, memory-evoking interfaces, enriching user engagement and accessibility while fostering a deeper connection with historical and cultural materials. By bridging personal experiences with archival memories, this system reimagines archives as living, interactive spaces that inspire curiosity and creativity. Although this work is specific to the subset of the RTS archive, at the core of this artwork is its ability to seamlessly translate any archival video material into diverse textual representations, and vice versa. Building upon this foundation, future endeavours can focus on the development of diverse interfaces and experiences aimed at delving into overarching themes within audiovisual archival art, historical narrative, and collaborative curation and creation using broader and various audiovisual materials for bespoke subjects and themes.

¹<https://github.com/tensorflow/tfjs-models/tree/master>

²<https://github.com/valadaptive/ntsc-rs>

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